

**SIMATS ENGINEERING
ASSESSMENT TOOL 1**

COURSE CODE: CSA5115

COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO5: *Develop the network security strategies based on the study of viruses, worms, firewall architectures, and IDS/IPS functionalities.CO (BL5, BL6)*

Assessment Tool Weightage: 40%

QUESTIONS: 5

TOTAL MARKS:

25

Annexure A

CONSTRAINT-BASED PROBLEM SOLVING

Code of Conduct:

I Nissanki Sri Venkata Lakshmi Kavya(192373036) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

N.S.V.L.Kavya

Annexure A

CONSTRAINT-BASED PROBLEM SOLVING

1. A financial institution must secure all web transactions using TLS, but legacy systems only support older SSL versions. Design a solution to ensure secure communication.

Constraints:

- A. Backward compatibility must be maintained for 20% legacy clients
- B. Sensitive data must not be transmitted using SSL
- C. System downtime must be minimal during migration
- D. Cost of implementation should be optimized

Proposed Solution

The financial institution should adopt a phased TLS migration strategy. The primary system should use modern TLS versions, while legacy clients should be isolated and gradually upgraded.

1. **Use TLS as the primary security protocol**
 - Configure all modern clients to use TLS 1.2 or TLS 1.3.
 - Disable SSL 2.0 and SSL 3.0 on the main production servers.
 - Use strong encryption algorithms and certificates.
2. **Maintain limited compatibility for legacy clients**
 - The 20% legacy clients can temporarily connect through a separate legacy gateway.
 - The gateway should prevent these clients from directly accessing sensitive transaction services.
 - Legacy connections should be restricted to non-sensitive operations such as basic account information or system updates.

3. Protect sensitive data

- Passwords, card details, banking information, transaction data, and other sensitive information must be processed only through TLS-secured connections.
- Legacy SSL clients should not be permitted to perform sensitive transactions.
- If a legacy client needs to perform a sensitive transaction, it should first be upgraded to a TLS-compatible version.

4. Perform migration in phases

- First identify all legacy clients and their SSL versions.
- Introduce the TLS infrastructure without immediately shutting down existing systems.
- Migrate modern clients first.
- Gradually upgrade the remaining 20% legacy clients.
- After migration, completely disable SSL support.

5. Minimize downtime

- Use load balancers and parallel TLS-enabled servers during migration.
- Perform configuration changes during low-traffic periods.
- Test TLS configurations before switching production traffic.
- Use a gradual client migration rather than a complete system replacement.

6. Optimize implementation cost

- Reuse existing servers and network infrastructure where possible.
- Upgrade only components that cannot support TLS.
- Use automated certificate management and centralized security policies.
- Prioritize critical and high-risk legacy systems first.

Final Decision

The institution should not continue using SSL for sensitive transactions. TLS 1.2/1.3 should become the standard communication protocol. A temporary isolated gateway can maintain limited compatibility for the 20% legacy clients while they are progressively upgraded.

Constraint Satisfaction

Constraint	Solution
20% legacy clients	Temporary isolated legacy gateway
No sensitive data over SSL	Sensitive transactions allowed only through TLS
Minimal downtime	Phased migration and parallel infrastructure
Cost optimization	Reuse infrastructure and upgrade only necessary systems

Conclusion: A phased migration to TLS provides strong security, maintains temporary backward compatibility, minimizes downtime, and controls implementation costs.

2. An e-commerce platform is considering implementing the SET protocol for payment security but currently uses TLS-based encryption. Analyze and propose a decision.

Constraints:

- A. Must support high transaction throughput (10,000 transactions/minute)
- B. Compatibility with modern browsers is required
- C. Implementation complexity should be minimized
- D. Regulatory compliance (PCI-DSS) must be ensured

Analysis

SET (Secure Electronic Transaction) was designed specifically for secure electronic payment transactions using digital certificates and cryptographic mechanisms. However, it is an older payment-security protocol and is not commonly supported by modern web browsers.

TLS, on the other hand, provides secure communication between the customer's browser and the e-commerce server and is widely supported by modern browsers.

Comparison

Factor	SET	TLS
Transaction throughput	More processing overhead	Efficient and scalable
Modern browser support	Very limited	Excellent
Implementation complexity	High	Lower
Payment-specific security	Strong	Strong when combined with secure payment processing
Scalability	More difficult	Highly scalable
PCI-DSS support	Can contribute to security	Widely used in PCI-DSS compliant systems
Current industry usage	Very limited	Industry standard

Proposed Solution

The e-commerce platform should continue using TLS rather than implementing SET.

1. Use TLS 1.2/1.3

1. Secure all communication between customers and the e-commerce platform.
2. Use strong cipher suites and valid digital certificates.
3. Disable obsolete SSL/TLS versions.

2. Support high transaction throughput

1. TLS can support large numbers of concurrent connections when properly configured.
2. Load balancing, connection reuse, and scalable server infrastructure can help achieve the requirement of **10,000 transactions per minute**.

3. Maintain browser compatibility

1. Modern browsers natively support TLS.
2. Customers do not need special SET software or certificates.
3. This provides a smoother shopping experience.

4. Reduce implementation complexity

1. The existing TLS infrastructure can be improved rather than replaced.
2. The organization avoids the additional complexity of deploying and maintaining SET certificates and payment-specific infrastructure.

5. Ensure PCI-DSS compliance

1. TLS should be combined with proper payment security controls.
2. Cardholder data should be protected during transmission and should not be unnecessarily stored.
3. Access controls, encryption, monitoring, vulnerability management, and regular security testing should also be implemented.

6. Use a secure payment gateway

1. The platform can use a PCI-DSS-compliant payment gateway or processor.
2. Tokenization can reduce the exposure of cardholder information.
3. This allows the e-commerce platform to focus on secure communication while the payment provider handles sensitive payment processing.

Final Decision

The e-commerce platform should not implement SET. It should continue with TLS

1.2/1.3 combined with a PCI-DSS-compliant payment gateway and appropriate security controls.

This approach provides:

- High transaction throughput
- Modern browser compatibility
- Lower implementation complexity
- Better scalability
- Support for PCI-DSS compliance

Constraint Satisfaction

Constraint	Solution
10,000 transactions/minute	Scalable TLS infrastructure, load balancing and connection reuse
Modern browser compatibility	TLS is natively supported
Minimum implementation complexity	Continue and strengthen existing TLS infrastructure
PCI-DSS compliance	TLS + secure payment gateway + required PCI-

Conclusion

TLS is the more suitable choice for the e-commerce platform because it is modern, widely supported, scalable, and easier to implement than SET. SET may provide specialized payment-security features, but its limited modern browser support and higher implementation complexity make it unsuitable for the given requirements.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
		accurately			
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided